Two-Period Model of Partial Sovereign Default

MPhil Thesis Progress

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1 The Model Environment

I present an endowment economy inhabited by three distinct agents: a representative household (the sovereign), a continuum of risk-neutral domestic import intermediaries (domestic banks) indexed by $i \in [0,1]$, and a risk-neutral international lender. The discrete-time model spans two periods, $t \in \{0,1\}$.

1.1 Primitives

The representative household has time-separable expected utility over consumption:

$$V_0 = u(C_0) + \beta\, \mathbb{E}_0\big[u(C_1)\big], \tag{1}$$

where u is strictly increasing, strictly concave, and satisfies Inada conditions, and $\beta \in (0,1)$ is the subjective discount factor.
The household produces the final consumption good C_t from domestic inputs m_t^d and foreign inputs m_t^f via a CES aggregator:

$$C_t = \big[\alpha(m_t^d)^\eta + (1-\alpha)(m_t^f)^\eta\big]^{1/\eta}, \qquad t \in \{0,1\}, \tag{2}$$

where $\alpha \in (0,1)$ is the Armington weight on domestic inputs and $\eta < 0$ governs the elasticity of substitution $\sigma = 1/(1-\eta) \in (0,1)$. The restriction $\eta < 0$ ensures the two inputs are strict gross complements: the household cannot substitute away from foreign inputs to avoid punitive price increases.
The household receives a per-period endowment y_t of domestic inputs. Period 1 endowment growth $g_1 \equiv y_1/y_0$ follows an AR(1) process in logs:

$$\log(g_1) = (1-\rho)\log(\mu_g) + \rho\log(g_0) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma_g^2). \tag{3}$$

There exists an exogenous risk-free gross world interest rate $R^* > 1$. Following Eaton and Gersovitz (1981), the international lender has access to a frictionless world capital market at this rate. As such, the lender will only participate in lending activities, both interperiod and intraperiod, which guarantee them a rate of return at least as high as this market rate.

1.2 Nominal Structure and Exchange Rate

Let P_t denote the domestic-currency price of domestic goods, and e_t the exchange rate, expressed as units of domestic currency per unit of foreign currency. The real exchange rate is defined as

$$RER_t \equiv \frac{e_t}{P_t}, \tag{4}$$

where foreign input prices are normalised to one. I assume that bonds within the domestic economy are issued in real-terms, representing a fixed quantity of domestic output, as to ensure the real net-of-default payoff for the household is invariant to domestic price levels.

Domestic input purchases are subject to a cash-in-advance constraint that pins down

domestic price levels:

$$MS_t = P_t \cdot m_t^d, \tag{5}$$

where MS_t denotes the domestic money supply.

Proposition 1 below establishes that, given neutrality assumptions, the real exchange rate reduces to $R\bar{E}R = 1$, with one unit of foreign input exchanging for one unit of domestic input on the (frictionless) world market. The remainder of the analysis proceeds in real terms.

1.3 Financial Instruments and the CSV Friction

The household issues one-period, non-contingent bonds b_1 at price q_0 in Period 0. Each bond promises one unit of the domestic numéraire at maturity. A fraction $\gamma \in (0, 1)$ of outstanding bonds is held by the domestic intermediary sector; the remainder $(1 - \gamma)$ is held by the international lender. The parameter γ is exogenous and institutional, reflecting a regulatory liquidity mandate requiring domestic banks to hold a fraction of outstanding sovereign bonds as eligible liquid assets. This captures the well-documented home bias in sovereign debt holdings across emerging market banking sectors (Gennaioli et al., 2014).
Upon realisation of the Period 1 endowment, the sovereign household chooses a continuous partial default $D \in [0, b_1]$, interpreted as a haircut on outstanding debt. As an important departure from Arellano (2008), the household does not internalise the effect of its default decision on the price of debt. Rather, default occurs in equilibrium as the household internalises the impact of default on import prices. The sovereign in this setting can be thought of as being subject to a coordination failure in the issuance of new debt (Goodhart et al., 2018).
Each intermediary is subject to an idiosyncratic productivity shock $\omega_{i,t}$, drawn independently each period from a log-normal distribution with mean one and variance σ_ω^2:

$$\log(\omega_{i,t}) \sim \mathcal{N}\left(-\tfrac{\sigma_\omega^2}{2},\, \sigma_\omega^2\right). \tag{6}$$

Let $F(\omega)$ and $f(\omega)$ denote the CDF and PDF of this distribution. Because the realised shock is private information to the intermediary, the financial contract between the intermediary and the international lender is subject to a costly state verification (CSV) friction, following Bernanke et al. (1999).
The CSV friction is characterised by two functions of the default threshold $\bar{\omega} \geq 0$ that appear throughout the analysis. The expected gross share of intermediary revenue accruing to the lender under the optimal debt contract is:

$$\Gamma(\bar{\omega}) \equiv \int_0^{\bar{\omega}} \omega \, dF(\omega) + \bar{\omega}\big[1 - F(\bar{\omega})\big], \tag{7}$$

and the expected revenue share from defaulting intermediaries is:

$$G(\bar{\omega}) \equiv \int_0^{\bar{\omega}} \omega \, dF(\omega), \tag{8}$$

so that $\mu G(\bar{\omega})$ is the expected deadweight auditing cost, where $\mu \in (0, 1)$ is the monitoring

cost parameter. These functions satisfy $\Gamma(0) = 0$, $\Gamma'(\bar{\omega}) = 1 - F(\bar{\omega}) > 0$, and $G'(\bar{\omega}) = \bar{\omega} f(\bar{\omega}) \geq 0$. Under the log-normal specification (6), the derivatives admit the closed forms:

$$\Gamma'(\bar{\omega}) = 1 - \Phi\left(\frac{\ln(\bar{\omega}) + \sigma_\omega^2/2}{\sigma_\omega}\right), \tag{9}$$

$$\mu G'(\bar{\omega}) = \frac{\mu}{\sigma_\omega}\,\varphi\left(\frac{\ln(\bar{\omega}) + \sigma_\omega^2/2}{\sigma_\omega}\right), \tag{10}$$

$$\Gamma''(\bar{\omega}) = -\frac{1}{\bar{\omega}\,\sigma_\omega}\,\varphi\left(\frac{\ln\bar{\omega} + \sigma_\omega^2/2}{\sigma_\omega}\right), \tag{11}$$

where Φ and φ denote the standard normal CDF and PDF. It is worth noting that Γ'' is strictly negative for all $\bar{\omega} > 0$.

1.4 Sequence of Events

The within-period timing is as follows.

Period 0:

(a) The household receives endowment y_0.

(b) The household issues bonds b_1 at price q_0; fraction γ purchased by intermediaries, remainder by the international lender.

(c) Intermediaries borrow working capital κ_0 from the international lender and intermediate foreign inputs m_0^f.

(d) Idiosyncratic shocks $\omega_{i,0}$ are realised; solvent intermediaries deliver inputs and distribute surplus Π_0 to the household.

(e) The household produces and consumes C_0.

Period 1:

(a) The aggregate endowment shock g_1 is realised, determining $y_1 = g_1 y_0$.

(b) The household chooses partial default $D \in [0, b_1]$.

(c) Intermediary net worth n_1 is updated.

(d) Intermediaries borrow working capital κ_1 subject to the CSV contract, determining the domestic price of foreign inputs $p_{m,1}$.

(e) Idiosyncratic shocks $\omega_{i,1}$ are realised; solvent intermediaries deliver inputs and distribute surplus Π_1 to the household.

(f) The household produces and consumes C_1.

2 Agents

2.1 Domestic Household

The representative household acts as both a consumer and a final-goods producer, aggregating domestic and foreign inputs via (2). It takes the internal price of foreign inputs, $p_{m,t}$ - the relative price expressed in units of domestic inputs - as given in each period.

Period 0. The household chooses domestic inputs m_0^d, foreign inputs m_0^f, and bond issuance b_1 to maximise (1) subject to:

$$m_0^d + p_{m,0}\, m_0^f = y_0 - b_0 + q_0 b_1 + \Pi_0, \tag{12}$$

where b_0 denotes pre-existing debt obligations and Π_t is a lump-sum dividend received from the domestic intermediary sector, defined below. The household acts as a price-taker in its own debt issuance.

Period 1. After y_1 is realised, the household chooses default $D \in [0, b_1]$ and inputs (m_1^d, m_1^f) to solve:

$$\max_{D\in[0,b_1],\, m_1^d\geq 0,\, m_1^f\geq 0} u(C_1), \tag{13}$$

subject to:

$$m_1^d + p_{m,1}(D) \cdot m_1^f = y_1 - (b_1 - D) + \Pi_1. \tag{14}$$

The import price $p_{m,1}$ depends on D through the intermediary balance sheet channel described below. In order for default to occur in equilibrium, the household must internalise this effect. Crucially, while the household internalises the dependence of $p_{m,1}$ on D, it treats the dividend Π_1 as a lump sum, that is, it does not internalise $\partial\Pi_1/\partial D$ when choosing default. This asymmetry reflects the fact that import prices are directly observable and contractually relevant to the household's purchasing decisions, whereas the dividend is a residual claim on aggregate intermediary surplus that the atomistic households do not perceive to depend on their own actions.

2.2 Domestic Intermediary

A continuum of ex-ante identical domestic intermediaries $i \in [0, 1]$ operate in a perfectly competitive market with free entry and exit to intermediate foreign inputs m^f in each period. Each intermediary purchases imports using a combination of its own net worth $n_{i,t}$ and working capital $\kappa_{i,t}$ borrowed from the international lender at gross intra-period rate $Z_{i,t}$, with repayment due within the same period. The intermediary's intra-period balance sheet is:

$$m_{i,t}^f = n_{i,t} + \kappa_{i,t}, \qquad t \in \{0, 1\}. \tag{15}$$

It is worth emphasising that $\kappa_{i,t}$ is an intra-period contract—borrowing and repayment occur within the same period t—distinct from the inter-period sovereign bond b_1 issued in Period 0 and maturing in Period 1. The international lender's opportunity cost is R^* for both contracts, but the two markets clear independently.

The intermediary's net worth is centred around an exogenous long-run level $\bar{n} > 0$ and depends on the period and the sovereign's default choice:

$$n_{i,0} = \bar{n} + \gamma b_0 - \gamma q_0 b_1, \tag{16}$$

$$n_{i,1} = \bar{n} + \gamma(b_1 - D). \tag{17}$$

Before delivering inputs to the household, intermediary i is subject to the idiosyncratic shock $\omega_{i,t}$, so that aggregate effective foreign inputs available for sale are:

$$M_t^f = \int_0^1 \omega_{i,t}\, m_{i,t}^f \, di. \tag{18}$$

The cutoff threshold $\bar{\omega}_{i,t}$ is the exact shock realisation at which the intermediary just breaks even:

$$\bar{\omega}_{i,t}\, p_{m,t}\, m_{i,t}^f = Z_{i,t}\, \kappa_{i,t} = Z_{i,t}\big(m_{i,t}^f - n_{i,t}\big). \tag{19}$$

If $\omega_{i,t} \geq \bar{\omega}_{i,t}$, the intermediary repays in full and retains the surplus. If $\omega_{i,t} < \bar{\omega}_{i,t}$, the intermediary defaults; the lender pays the auditing cost, destroying a fraction μ of realised assets and recovering $(1-\mu)\omega_{i,t}\, p_{m,t}\, m_{i,t}^f$.
Each intermediary chooses $m_{i,t}^f$ to maximise expected profits:

$$\max_{m_{i,t}^f} \big[1 - \Gamma(\bar{\omega}_{i,t})\big]\, p_{m,t}\, m_{i,t}^f, \tag{20}$$

yielding the first-order condition:

$$\big[1 - \Gamma(\bar{\omega}_{i,t})\big]\, p_{m,t}\, m_{i,t}^f = \Gamma'(\bar{\omega}_{i,t})\, Z_{i,t}\, n_{i,t}. \tag{21}$$

Unlike the bilateral optimal contract in Christiano et al. (2014), intermediaries here take $Z_{i,t}$ as given in a competitive credit market. They do not internalise the marginal auditing costs when choosing their scale of intermediation, generating a steeper external finance premium for a given level of net worth. This departure reflects the competitive and fragmented nature of the domestic banking sector.

Intermediary dividends. All intermediaries are owned by the representative household, who acts as the residual claimant on any surplus they generate. Once the idiosyncratic shock is realised and obligations to the international lender are settled, solvent intermediaries distribute their surplus revenue as a lump-sum dividend. The per-unit surplus of intermediary i, net of working-capital repayment, is:

$$\pi_{i,t} = \begin{cases} (\omega_{i,t} - \bar{\omega}_t)\, p_{m,t}\, m_{i,t}^f & \text{if } \omega_{i,t} \geq \bar{\omega}_t, \\ 0 & \text{otherwise.} \end{cases} \tag{22}$$

Aggregating across the continuum of intermediaries and applying the law of large numbers:

$$\Pi_t = \int_{\bar{\omega}_t}^{\infty} (\omega - \bar{\omega}_t)\, p_{m,t}\, M_t^f \, dF(\omega) = \big[1 - \Gamma(\bar{\omega}_t)\big]\, p_{m,t}\, M_t^f. \tag{23}$$

This is the aggregate dividend Π_t that enters the household's budget constraints (12)–(14).

2.3 International Lender

The international lender is risk-neutral, infinitely funded, and prices all contracts at the risk-free rate R^*.

Working capital market. The lender's zero-profit participation constraint equates the expected payoff to the opportunity cost:

$$\left[\int_{\bar{\omega}_{i,t}}^{\infty} \bar{\omega}_{i,t}\, dF(\omega) + (1-\mu)\int_{0}^{\bar{\omega}_{i,t}} \omega\, dF(\omega)\right] p_{m,t}\, m_{i,t}^{f} = R^* \kappa_{i,t}. \tag{24}$$

Using the CSV functions from Section 1.3, this simplifies to:

$$\big[\Gamma(\bar{\omega}_{i,t}) - \mu G(\bar{\omega}_{i,t})\big]\, p_{m,t}\, m_{i,t}^{f} = R^* \big(m_{i,t}^{f} - n_{i,t}\big). \tag{25}$$

Intertemporal bond market. The lender prices household debt at the risk-neutral expected recovery rate:

$$q_0(y_0, b_1) = \frac{1}{R^*}\, \mathbb{E}_0\left[1 - \frac{D(y_1, b_1)}{b_1}\right]. \tag{26}$$

3 Equilibrium Definitions

3.1 Aggregation and Market Clearing

Because all intermediaries are ex-ante identical, in any symmetric equilibrium they hold the same net worth ($n_{i,t} = N_t$), purchase the same raw imports ($m_{i,t}^f = M_t^f$), borrow the same working capital ($\kappa_{i,t} = K_t$), and face the same default threshold ($\bar{\omega}_{i,t} = \bar{\omega}_t$). Henceforth all intermediary-level variables are written without the i subscript.

Three markets must clear in equilibrium. The sovereign debt market requires that total issuance equals total demand:

$$b_1 = \gamma b_1 + (1-\gamma) b_1. \tag{27}$$

The foreign inputs market requires that the household's demand for effective foreign inputs equals aggregate intermediary supply net of auditing losses:

$$m_t^f = M_t^f \big[1 - \mu G(\bar{\omega}_t)\big]. \tag{28}$$

The domestic inputs market clears by Walras' law given the first two conditions and the household budget constraint.

Aggregating across agents and imposing market clearing, the economy's balance-of-payments constraints are:

$$m_0^d + R^* M_0^f - (R^*-1) N_0 = y_0 + \bar{n} - (1-\gamma)(b_0 - q_0 b_1), \tag{29}$$

$$m_1^d + R^* M_1^f - (R^*-1) N_1 = y_1 + \bar{n} - (1-\gamma)(b_1 - D). \tag{30}$$

3.2 Equilibrium Definition

Definition 1 (Competitive Equilibrium)**.** Given initial debt b_0, long-run net worth $\bar{n}$, exogenous global parameter R^* and realised aggregate endowments (y_0, y_1), a competitive equilibrium is defined as a collection of prices $\{q_0,\, p_{m,0},\, p_{m,1},\, Z_0,\, Z_1\}$, thresholds $\{\bar{\omega}_0,\, \bar{\omega}_1\}$, and allocations $\{C_t,\, m_t^d,\, m_t^f,\, M_t^f,\, b_1,\, D,\, N_t,\, K_t,\, \Pi_t\}_{t=0,1}$ such that:

(i) The representative household optimises lifetime utility, choosing (m_t^d, m_t^f, b_1, D) given prices $(q_0, p_{m,0}, p_{m,1})$;

(ii) Each domestic intermediary selects $m_{i,t}^f$ to maximise expected profit (20), given the lending rate Z_t;

(iii) The international lender prices sovereign bonds via (26) and the working capital zero-profit condition (25) holds;

(iv) The sovereign debt, foreign input, and domestic input markets clear;

(v) Net worth satisfies (16)–(17), working capital is consistent: $K_t = M_t^f - N_t$, and aggregate dividends satisfy (23).

3.3 Nominal Neutrality

Under the nominal structure defined in Section 1.2, the following result establishes that the real equilibrium is self-contained:

Lemma 1 (Nominal Neutrality)**.** *Suppose that:*

(i) sovereign bonds are denominated in units of the domestic good, so that the real payoff $(b_1 - D)$ is independent of P_t;

(ii) domestic goods prices are fully flexible; and

(iii) the real exchange rate is constant: $RER_t = R\bar{E}R$ for all t.

Then the real allocation $\{C_t,\, m_t^d,\, m_t^f,\, M_t^f,\, b_1,\, D,\, \bar{\omega}_t,\, Z_t,\, \Pi_t\}_{t=0,1}$ and the real prices $\{q_0,\, p_{m,0},\, p_{m,1}\}$ are determined by the system of equations (2)–(45) independently of the domestic money supply MS_t, the nominal exchange rate e_t, and the domestic price level P_t. The nominal variables are determined residually by:

$$P_t = \frac{MS_t}{m_t^d}, \qquad e_t = P_t \cdot R\bar{E}R, \tag{31}$$

where $R\bar{E}R = 1$.

Proof. Express each agent's budget constraint and optimality condition in nominal terms, then deflate by P_t. The household's Period 1 nominal budget constraint is

$$P_1 \cdot m_1^d + P_{m,1}^{\text{nom}} \cdot m_1^f = P_1 \cdot y_1 - P_1 \cdot (b_1 - D) + P_1 \cdot \Pi_1,$$

where $P_{m,1}^{\text{nom}}$ is the nominal domestic-currency price of foreign inputs. Dividing by P_1 and letting $p_{m,1} \equiv P_{m,1}^{\text{nom}}/P_1$ recovers (14). The relative price $p_{m,t}$ thus captures the full real

cost of foreign inputs to the domestic household, inclusive of both the underlying terms of trade and the friction-based markup, detailed in Section 1.3.
The same exercise can be conducted for Period 0 equilibrium, recovering the stated real system by deflation as above.
The intermediary balance sheet in nominal terms is $e_t \cdot m^f_{i,t} = P_t \cdot n_{i,t} + e_t \cdot \kappa_{i,t}$. Dividing by P_t and noting $e_t/P_t = 1$ under assumption (iii) yields (15).
The CSV functions $\Gamma(\bar{\omega})$ and $G(\bar{\omega})$ are unit-free ratios, so the threshold condition (19), the lender's zero-profit condition (24), and the import price equation (37) are invariant to the choice of numéraire.
Since every equilibrium condition in the real system is recovered by deflation, the real allocation is independent of MS_t. Given the equilibrium values of m^d_t, the price level follows from (5) and the nominal exchange rate from (31).

□

Let i_t denote the domestic nominal interest rate. Under satisfaction of the uncovered interest rate parity condition, the nominal rate can be expressed as

$$(1 + i_t) = R^* \, \mathbb{E}_t \left[\frac{e_{t+1}}{e_t} \right]. \tag{32}$$

Under the assumption of nominal neutrality, with domestic prices being pinned down by (5), this becomes

$$(1 + i_t) = R^* \, \mathbb{E}_t \left[\frac{MS_{t+1}/m^d_{t+1}}{MS_t/m^d_t} \right]. \tag{33}$$

The domestic nominal rate therefore depends on monetary policy and the real allocation of domestic inputs, but is determined residually and does not feed back into the real equilibrium.
Following the Fisher equation, the domestic real interest rate R_t is expressed as

$$R_t = (1 + i_t) \frac{P_t}{\mathbb{E}_t[P_{t+1}]} = R^*, \tag{34}$$

with the domestic rate equalising each period to the time-invariant world gross rate R^*.

4 Optimality Conditions

4.1 The Financial Friction

Three key relationships emerge from the intermediary and lender optimality conditions. Aggregating across intermediaries and dividing the threshold condition (19) by the zero-profit condition (25), the equilibrium lending rate satisfies:

$$Z_t = R^* \cdot \underbrace{\frac{\bar{\omega}_t}{\Gamma(\bar{\omega}_t) - \mu G(\bar{\omega}_t)}}_{\text{external finance premium}}. \tag{35}$$

The ratio Z_t/R^* exceeds unity whenever $\bar{\omega}_t > 0$ and is increasing in $\bar{\omega}_t$.

Dividing the intermediary FOC (21) by the zero-profit condition (25) and substituting (35), the threshold $\bar{\omega}_t$ is pinned down by the leverage ratio alone:

$$1 - \Gamma(\bar{\omega}_t) = \Gamma'(\bar{\omega}_t)\,\bar{\omega}_t \cdot \frac{N_t}{M_t^f - N_t}. \tag{36}$$

For given M_t^f, a decrease in N_t raises leverage and increases $\bar{\omega}_t$.
Solving the zero-profit condition (25) for $p_{m,t}$ yields the import price equation:

$$p_{m,t} = R^* \cdot \underbrace{\left(1 - \frac{N_t}{M_t^f}\right)}_{\text{leverage}} \cdot \underbrace{\frac{1}{\Gamma(\bar{\omega}_t) - \mu G(\bar{\omega}_t)}}_{\text{CSV friction}}. \tag{37}$$

I assume intermediaries are sufficiently well capitalised in Period 0 that the external finance premium is negligible, so $p_{m,0} \approx R^*$.

4.2 Household Intratemporal Optimality

The first-order conditions with respect to m_t^d and m_t^f from the CES aggregator (2) yield the standard intratemporal demand function:

$$m_t^f = m_t^d \left(\frac{1-\alpha}{\alpha\, p_{m,t}}\right)^{\frac{1}{1-\eta}}, \tag{38}$$

and the shadow value of income:

$$\lambda_t = u'(C_t)\,\frac{\partial C_t}{\partial m_t^d} = u'(C_t)\,\alpha\left(\frac{C_t}{m_t^d}\right)^{1-\eta}. \tag{39}$$

4.3 Optimal Default

Lemma 2 (Optimal Default Policy)**.** *The household's first-order condition with respect to $D \in [0, b_1]$ is:*

$$\frac{\partial \mathcal{L}_1}{\partial D} = \lambda_1 \left(1 - m_1^f \frac{\partial p_{m,1}}{\partial D}\right). \tag{40}$$

The marginal benefit of an additional unit of default is one unit of budget relief. The marginal cost is the induced increase in the domestic import price, weighted by import demand. The full Karush–Kuhn–Tucker conditions are:

$$\begin{cases} D = 0 & \text{if } 1 \le m_1^f \left.\dfrac{\partial p_{m,1}}{\partial D}\right|_{D=0} \quad \text{(no default)}, \\ 1 = m_1^f \dfrac{\partial p_{m,1}}{\partial D} & \text{if } D \in (0, b_1) \quad \text{(partial default)}, \\ D = b_1 & \text{if } 1 \ge m_1^f \left.\dfrac{\partial p_{m,1}}{\partial D}\right|_{D=b_1} \quad \text{(full default)}. \end{cases} \tag{41}$$

At an interior solution, the household equates the marginal budgetary gain from one extra unit of default with the marginal cost through the external finance premium.

4.4 Transmission of Default to Import Prices

The mechanism by which sovereign default raises import prices can be traced through the intermediary balance sheet. From the net-worth equation (17):

$$\frac{\partial p_{m,1}}{\partial D} = -\gamma \frac{\partial p_{m,1}}{\partial N_1}. \tag{42}$$

An increase in D erodes intermediary net worth, raising leverage, pushing up the default threshold, increasing the external finance premium, and passing through to the import price:

$$D \uparrow \xrightarrow{(17)} N_1 \downarrow \xrightarrow{(36)} \bar{\omega}_1 \uparrow \xrightarrow{(35)} \frac{Z_1}{R^*} \uparrow \xrightarrow{(37)} p_{m,1} \uparrow \xrightarrow{(38)} m_1^f \downarrow \Longrightarrow C_1 \downarrow .$$

Differentiating (37) with respect to N_1 at fixed M_1^f:

$$\frac{\partial p_{m,1}}{\partial N_1} = R^* \cdot \frac{-\frac{1}{M_1^f}\left[\Gamma(\bar{\omega}_1) - \mu G(\bar{\omega}_1)\right] - \left(1 - \frac{N_1}{M_1^f}\right)\left[\Gamma'(\bar{\omega}_1) - \mu G'(\bar{\omega}_1)\right]\frac{\partial \bar{\omega}_1}{\partial N_1}}{\left[\Gamma(\bar{\omega}_1) - \mu G(\bar{\omega}_1)\right]^2}. \tag{43}$$

This is strictly negative. The first term in the numerator captures the direct reduction in leverage; the second captures the indirect effect through the lowering of $\bar{\omega}_1$, which reduces expected auditing costs. The denominator $[\Gamma - \mu G]^2$ is the source of convexity in the default penalty: as net worth falls in default and $\bar{\omega}$ rises, $\Gamma - \mu G$ shrinks, causing the denominator to contract and the magnitude of $\partial p_{m,1}/\partial N_1$ to accelerate.
To complete the derivation, implicitly differentiating (36) with respect to N_1 at fixed M_1^f:

$$\frac{\partial \bar{\omega}_1}{\partial N_1} = \frac{-\Gamma'(\bar{\omega}_1)\,\bar{\omega}_1 \cdot \frac{M_1^f}{(M_1^f - N_1)^2}}{\Gamma'(\bar{\omega}_1) + \left[\Gamma''(\bar{\omega}_1)\,\bar{\omega}_1 + \Gamma'(\bar{\omega}_1)\right]\frac{N_1}{M_1^f - N_1}}, \tag{44}$$

which is strictly negative provided the second-order conditions of the intermediary's problem are satisfied. Higher net worth lowers the default threshold. The numerator is negative because $\Gamma' > 0$ and $\bar{\omega} > 0$; the denominator is positive by the second-order conditions.

Proposition 1 (Convexity of the Default Penalty)**.** *The penalty to the Representative Household of default is convex in D: a linear increase in sovereign default triggers a more-than-proportional increase in the domestic import price $p_{m,1}$. This follows from two reinforcing effects as leverage increases:*

(i) $|\partial \bar{\omega}_t / \partial N_t|$ grows, since the numerator of (44) expands via the $(M_t^f - N_t)^2$ term while the denominator contracts;

(ii) $[\Gamma(\bar{\omega}_t) - \mu G(\bar{\omega}_t)]^2$ in the denominator of (43) shrinks as $\bar{\omega}_t$ rises.

Together, $\partial p_{m,1}/\partial D$ is increasing in D.

Proposition 2 (Default increases in the ratio of debt-to-endowment)**.** *Assume an interior default equilibrium where $D \in (0, b_1)$ and the second-order condition for utility maximization is strictly satisfied. The optimal sovereign default volume $D(y_1, b_1)$ is:*

(i) Strictly decreasing in the realised Period 1 endowment y_1.

(ii) Strictly increasing in the Period 1 debt issuance b_1.

Consequently, the optimal haircut is governed by the household's capacity to pay, increasing alongside the debt-to-endowment ratio b_1/y_1.

Proof. Let $F(D; y_1, b_1)$ denote the first-order condition for optimal default (41), under an interior equilibrium. Also, let $W_1 \equiv y_1 - (b_1 - D) + \Pi_1$ denote the household's Period 1 wealth. By Proposition 1, the derivative of $F(.)$ the with respect to D is strictly negative. Applying the Implicit Function Theorem, the comparative statics of optimal default with respect to the state variable y_1 is given by

$$\frac{\partial F(D; y_1, b_1)}{\partial y_1} = -\left(\frac{\partial m_1^f}{\partial W_1}\frac{\partial W_1}{\partial y_1}\right)\frac{\partial p_{m,1}}{\partial D}.$$

Since $\frac{\partial W_1}{\partial y_1} = 1$, $\frac{\partial p_{m,1}}{\partial D} > 0$ and, as $\eta < 0$, $\frac{\partial m_1^f}{\partial W_1} > 0$, it follows that $\frac{\partial F(D;y_1,b_1)}{\partial y_1} < 0$. Therefore $\frac{\partial D}{\partial y_1} < 0$.
Differentiating $F(.)$ with respect to b_1 yields

$$\frac{\partial F(D; y_1, b_1)}{\partial b_1} = -\left(\frac{\partial m_1^f}{\partial W_1}\frac{\partial W_1}{\partial n_1}\right)\frac{\partial p_{m,1}}{\partial D} + m_1^f\left(\frac{\partial}{\partial b_1}\frac{\partial p_{m,1}}{\partial D}\right).$$

Due to the construction of claims made in Period 0, $\frac{\partial W_1}{\partial b_1} = -1$, in turn guaranteeing $\frac{\partial m_1^f}{\partial b_1} < 0$. As b_1 enters positively into the Domestic Intermediary's Period 1 net-worth (14), from (44) we can see that $\frac{\partial}{\partial b_1}\frac{\partial p_{m,1}}{\partial D} < 0$. These two derivatives mean that $\frac{\partial F}{\partial b_1} > 0$, implying $\frac{\partial D}{\partial b_1} > 0$. □

4.5 Intertemporal Optimality

Lemma 3 (Euler Equation)**.** *The household's first-order condition with respect to b_1, applying the envelope theorem to V_1, yields:*

$$\lambda_0\, q_0 = \beta\, \mathbb{E}_0\left[\lambda_1\left(1 + \gamma\, m_1^f \frac{\partial p_{m,1}}{\partial N_1}\right)\right]. \tag{45}$$

Proof. By the envelope theorem, the marginal value of an additional unit of debt carried into Period 1 is:

$$\frac{\partial V_1}{\partial b_1} = \lambda_1\left(-1 - m_1^f \frac{\partial p_{m,1}}{\partial b_1}\right).$$

Since $N_1 = \bar{n} + \gamma(b_1 - D)$ and default D is taken as given at the time of issuance:

$$\frac{\partial p_{m,1}}{\partial b_1} = \gamma \frac{\partial p_{m,1}}{\partial N_1}.$$

Note the sign: higher b_1 raises intermediary net worth (since banks hold γb_1 in bonds), so $\partial p_{m,1}/\partial b_1 < 0$. Combining with the Period 0 first-order condition $\lambda_0 q_0 = \beta\, \mathbb{E}_0[-\partial V_1/\partial b_1]$ and simplifying yields (45). □

The left-hand side of (45) is the marginal utility cost of purchasing one unit of debt at price q_0 in Period 0. The right-hand side comprises two components: the direct repayment of one unit of resources, and the improvement in Period 1 import prices from the additional net worth that γ units of bond holdings deliver to intermediaries. Since $\partial p_{m,1}/\partial N_1 < 0$, the term $\gamma m_1^f(\partial p_{m,1}/\partial N_1)$ is negative. Higher debt issuance strengthens bank balance sheets but also increases the household's future repayment burden. In equilibrium, the household balances the Period 0 consumption benefit of additional borrowing against both the direct repayment cost and the indirect effect on import prices.

5 Characterisation of Equilibrium

5.1 Period 1 Equilibrium

Take as given the states (y_1, b_1) and the parameter R^*. Define the following shorthand, which depend only on $(D, \bar{\omega}_1, M_1^f)$ and parameters:

$$\begin{aligned} N_1(D) &\equiv \bar{n} + \gamma(b_1 - D), \\ \ell_1(D, M_1^f) &\equiv 1 - \frac{N_1(D)}{M_1^f}, \\ \Psi(\bar{\omega}_1) &\equiv \Gamma(\bar{\omega}_1) - \mu G(\bar{\omega}_1), \\ p_{m,1}(D, \bar{\omega}_1, M_1^f) &= R^* \cdot \frac{\ell_1}{\Psi(\bar{\omega}_1)}. \end{aligned}$$

The Period 1 equilibrium values of $(D, \bar{\omega}_1, M_1^f)$ solve the following nonlinear system.

Leverage–threshold condition [from (36) and (17)]:

$$1 - \Gamma(\bar{\omega}_1) = \Gamma'(\bar{\omega}_1)\,\bar{\omega}_1 \cdot \frac{N_1(D)}{M_1^f - N_1(D)}. \tag{46}$$

Foreign input market clearing [from (38), (14), (28), and (23)]:

$$M_1^f\left[1 - \mu G(\bar{\omega}_1)\right]\left[\left(\frac{\alpha\, p_{m,1}}{1-\alpha}\right)^{\sigma} + p_{m,1}\right] = y_1 - (b_1 - D) + \left[1 - \Gamma(\bar{\omega}_1)\right] p_{m,1}\, M_1^f. \tag{47}$$

Interior default condition [from (41), (42), and (43)]:

$$\Psi(\bar{\omega}_1)^2 = \gamma R^* M_1^f\left[1 - \mu G(\bar{\omega}_1)\right]\left(\frac{\Psi(\bar{\omega}_1)}{M_1^f} + \ell_1 \Psi'(\bar{\omega}_1)\left|\frac{\partial \bar{\omega}_1}{\partial N_1}\right|\right), \tag{48}$$

where

$$\frac{\partial \bar{\omega}_1}{\partial N_1} = \frac{-\Gamma'(\bar{\omega}_1)\,\bar{\omega}_1\, M_1^f}{(M_1^f - N_1)\Big[\Gamma'(\bar{\omega}_1)(M_1^f - N_1) + N_1\big[\Gamma''(\bar{\omega}_1)\,\bar{\omega}_1 + \Gamma'(\bar{\omega}_1)\big]\Big]}.$$

At corner solutions, (48) is replaced by $D = 0$ or $D = b_1$ with the corresponding inequality from the KKT conditions (41).

All other Period 1 variables are determined residually: N_1 from (17), $p_{m,1}$ from (37), m_1^f from (28), Π_1 from (23), m_1^d from (14), C_1 from (2), Z_1 from (35), and $K_1 = M_1^f - N_1$.

5.2 Period 0 Equilibrium

Given Period 1 policy functions $D(y_1, b_1)$, and the associated Period 1 equilibrium mappings, the Period 0 equilibrium values of (b_1, q_0) solve the following system.

Euler equation [from (45)]:

$$\lambda_0 \, q_0 = \beta \, \mathbb{E}_0 \left[\lambda_1 \left(1 + \gamma \, m_1^f \frac{\partial p_{m,1}}{\partial N_1} \right) \right], \tag{49}$$

where λ_t is derived in (39).

Intertemporal bond pricing equation [from (26)]:

$$q_0 = \frac{1}{R^*} \, \mathbb{E}_0 \left[1 - \frac{D(y_1, b_1)}{b_1} \right]. \tag{50}$$

Given a solution, the remaining Period 0 objects are recovered as: N_0 from (16), m_0^f from (38), m_0^d from (12), C_0 from (2), Π_0 from (23), and $K_0 = M_0^f - N_0$. Under the maintained assumption that Period 0 intermediaries are sufficiently capitalised, $\bar{\omega}_0$ and Z_0 are pinned down by (36) and (35) at negligible values of the external finance premium.

Ruling out no-trade equilibrium For the rest of the paper, attention will be restricted to the non-trivial interior partial default equilibrium, in which $q_0 > 0$ and $D \in (0, b_1)$;

Assumption 1. Parameters $(\bar{n}, \gamma, \sigma_\omega, \mu, \eta, \alpha, R^*)$ and states (y_1, b_1) are such that the KKT conditions (41) admit an interior solution $D \in (0, b_1)$.

The model admits a second class of equilibria in which the international lender anticipates full default, sets $q_0 = 0$, and the household is indifferent between not borrowing and borrowing for free with the intention of defaulting upon all claims, as identified by Dubey et al. (2005) in models of default with non-contingent debt. The two-period setting presented in this model, without a continuation value from market access, means that the household is exactly indifferent between these two equilibria. I choose to omit the analysis of such an equilibrium, following the practice of Arellano (2008), Peiris et al. (2025) and others, to focus upon the borrowing, default, and intermediary channels that are of primary interest to this thesis.

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